Academic Book Recommender System for University Students Using Content-Based Filtering

Graduation Project – Bachelor of Computer Science

Department of Computer Science

Majmaah University



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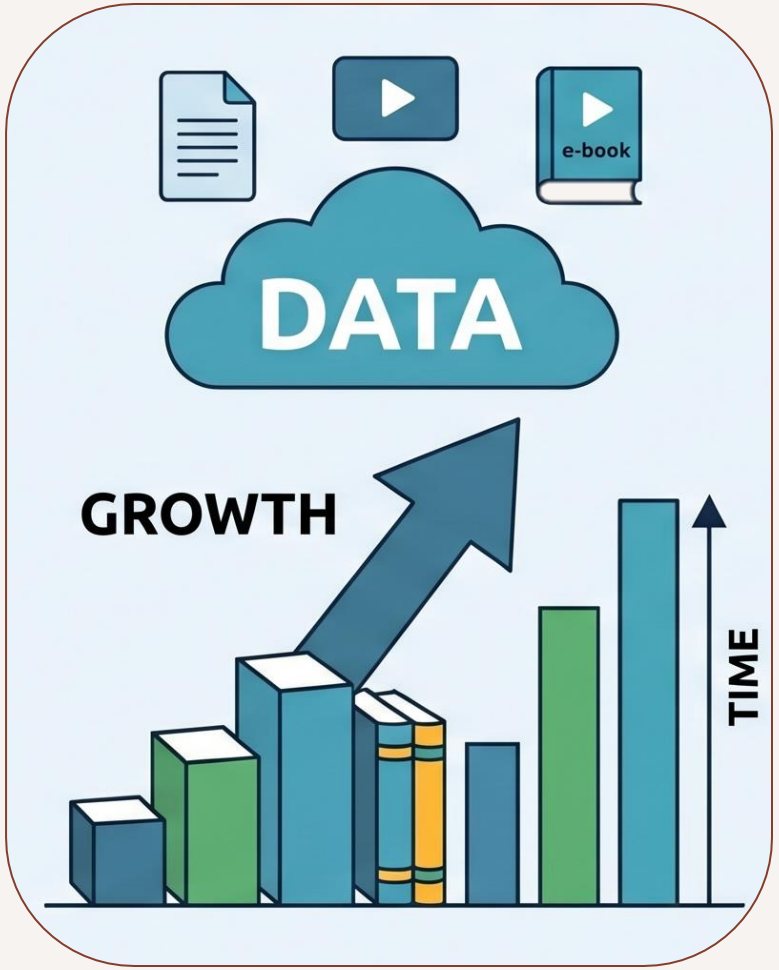
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1. Introduction & Background

Project Background

- Proliferation of digital libraries increases resource search friction.
- Traditional university catalogs offer rigid, non-personalized exploration paths.
- Al-Warraq aims to automate scholarly discovery by matching specific student profiles directly with highly relevant domain materials.



2. Introduction & Motivation

The Academic Shift: Massive expansion of scholarly digital libraries increases resource search friction.

The Goal: Automate resource discovery within narrow computing domains (e.g., Computer Science).

Al-Warraq Solution: An intelligent web platform that matches individual student profiles and specific disciplines directly with relevant book metadata.

3. Problem Statement

Information Overload: The massive explosion of digital book data makes it extremely difficult for students to find relevant academic resources.

Rigid Search Systems: Traditional libraries rely on basic keyword searches, ignoring semantic context and lacking personalization.

Wasted Time & Effort: Students spend excessive hours manually filtering through databases, which negatively impacts their academic productivity.

The Solution

Developing an intelligent, content-based recommendation system to provide personalized book suggestions and eliminate manual search friction.

3. Project Scope & Boundaries

In-Scope Features:

- Core NLP-driven recommendation engine.
- Interactive AI Conversational Librarian.
- User Profile creation & Academic interest tuning.
- Persistent database tier for saved favorites.

Out-of-Scope Elements:

- Standalone Mobile Apps (System is purely a web prototype).
- Advanced Hybrid filtering (Targeted for initial deployment stability).
- Cross-Cultural Recommendations in Multiple Languages

Expected Outcomes

1. **Experimental Platform:** Fully integrated Python/Flask web app.
2. **Data Sufficiency:** Ingests 5K+ records showing complete metadata (Title, Author, Category).
3. **Course Alignment:** Directly addresses student specialized disciplines.

4. System Requirements (FRs & NFRs)

Functional Requirements (FR):

1. User profile interest tagging and customization.
2. TF-IDF vector extraction and Cosine Similarity calculation.
3. Persistent storage for individual user favorites list.
4. Automated Chatbot dialogue answering queries based on book descriptions or knowledge.

Non-Functional Requirements (NFR):

Performance: Search operations must resolve in under 150ms.

Scalability: System must handle up to 15 concurrent users stably.

Availability: Target of 95% total server availability.

Safety: Strictly deterministic AI responses (0.1 Temperature).

5. Base Scope vs Advanced Improvements

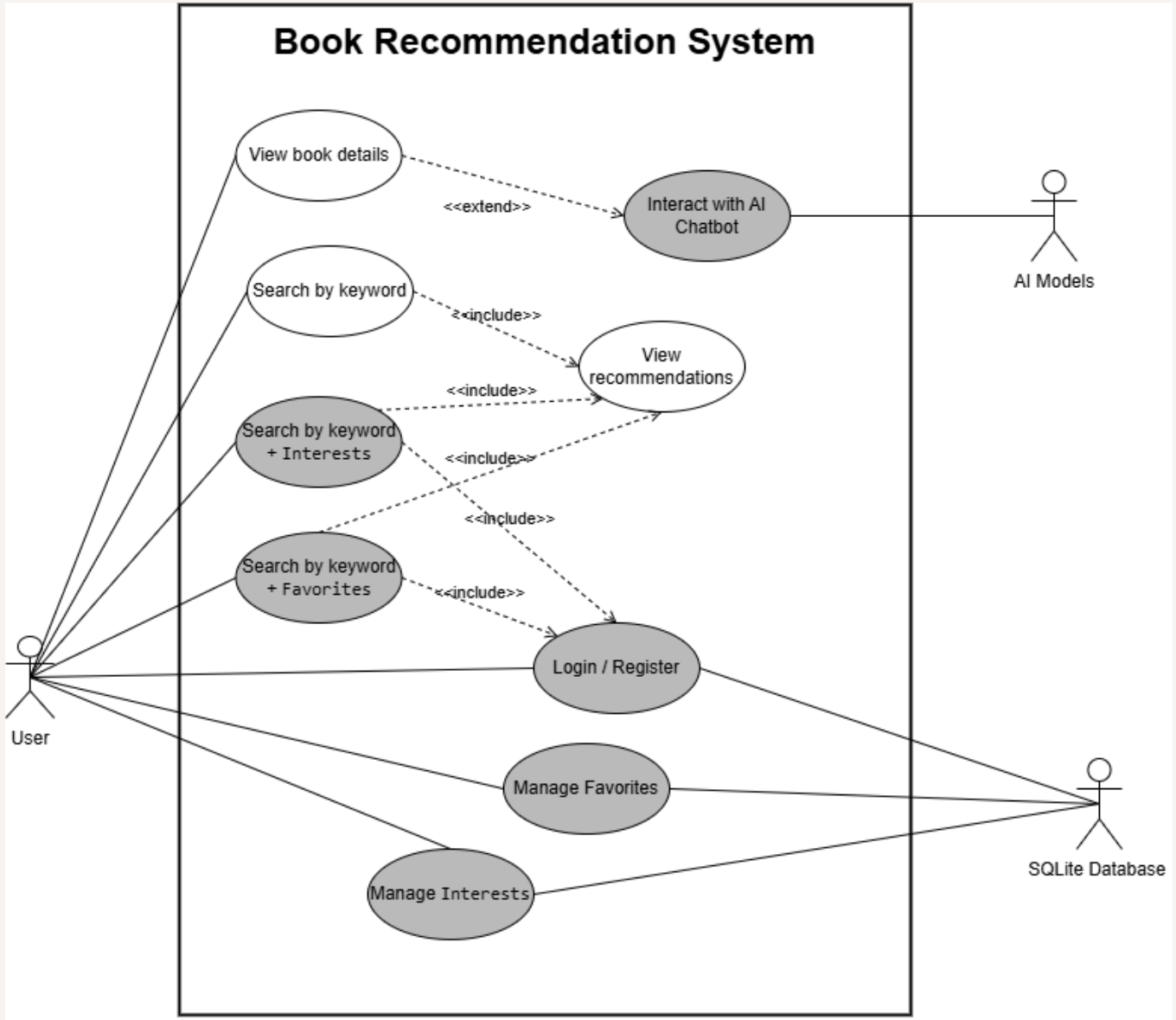
Standard Expected Scope:

- Basic keyword mapping and lookup.
- Simple static data storage and retrieval.

Al-Warraq Engineering Enhancements (Beyond the Scope):

- **Multi-Phase Scoring:** Combining raw queries with profile interests and saved favorites weights.
- **NLP Query Resilience:** Real-time typo correction and acronym expansion.
- **In-Memory Optimization:** Loading pre-computed matrix configurations entirely in RAM .
- **High-Availability AI Infrastructure:** Automated multi-cloud fallback routing.

6. System Architecture & Tech Stack



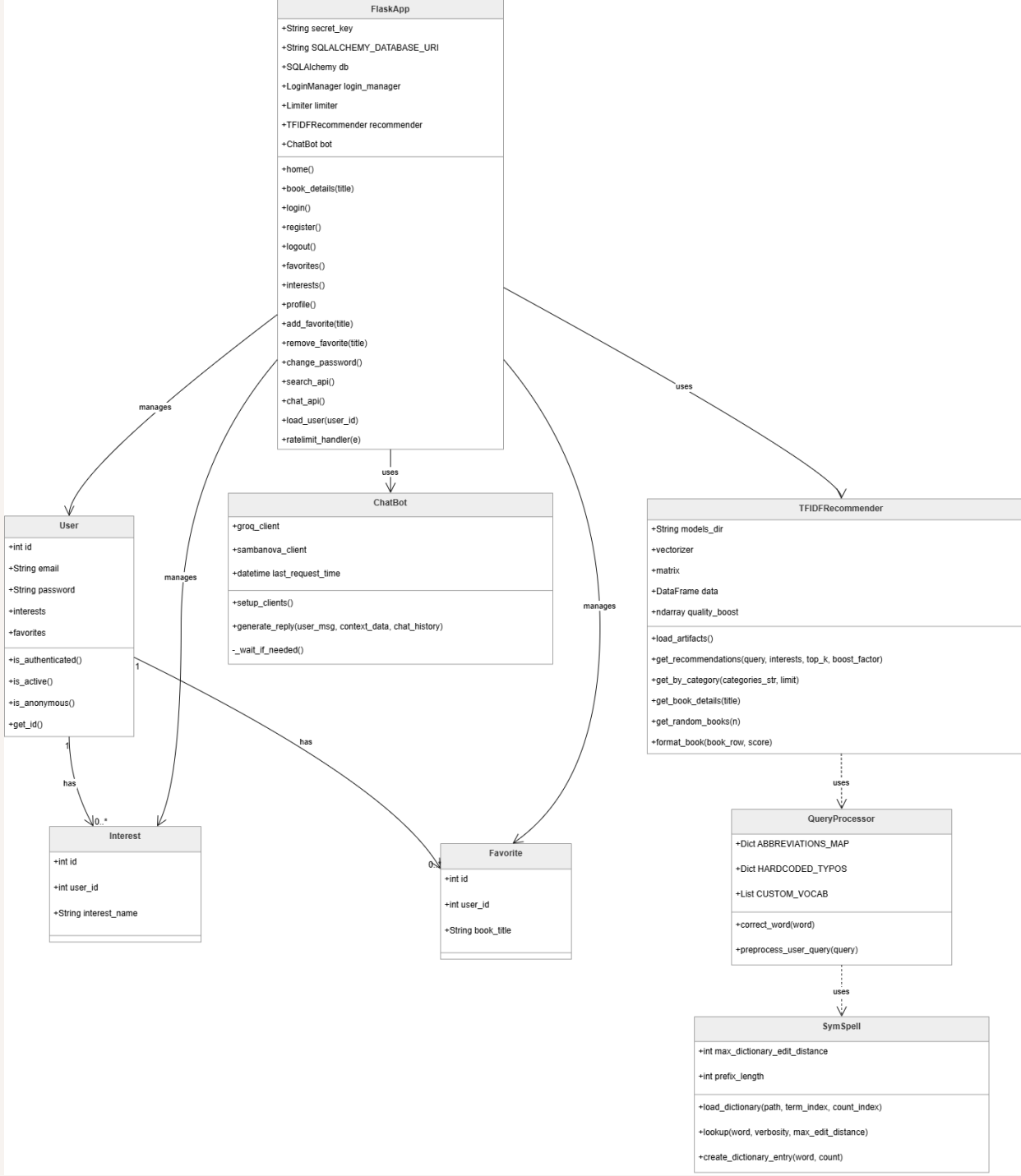
This diagram highlights the system's key enhancements, specifically the personalized search features and the AI chatbot integration.

6. System Architecture & Tech Stack

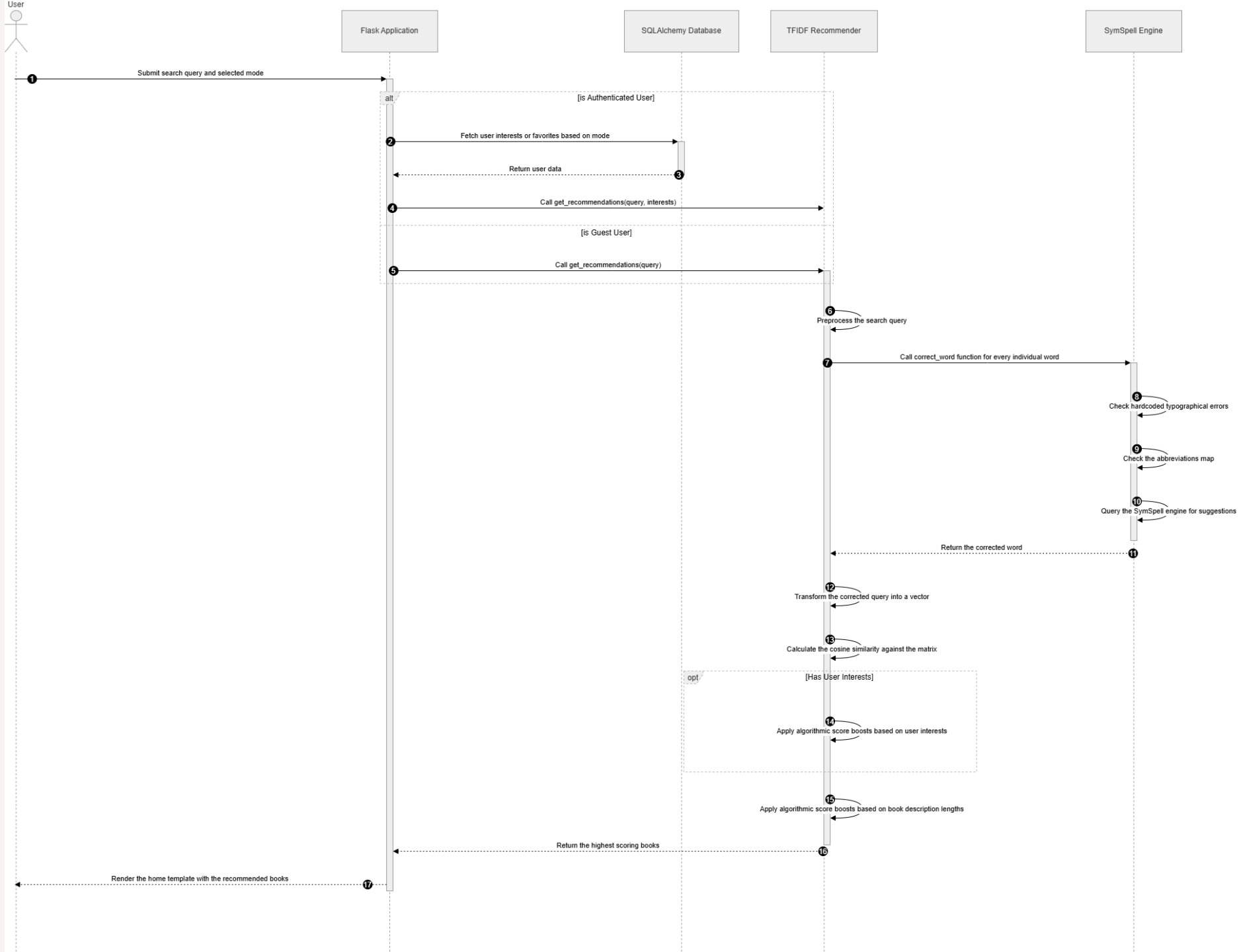
System Architecture

•Infrastructure:

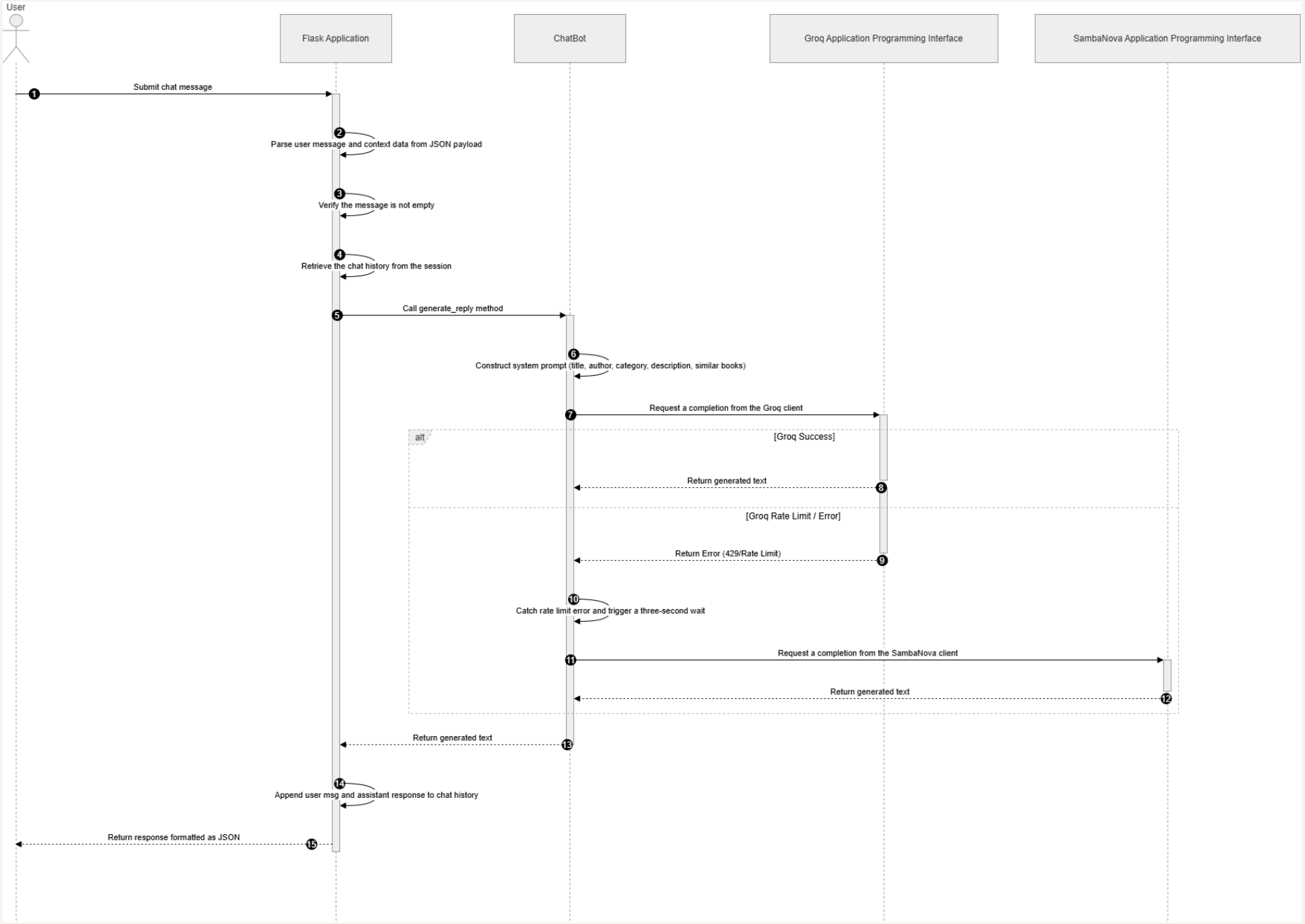
- **Backend:** Python / Flask Framework
- **Database:** SQLite with SQLAlchemy
- **NLP Engine:** Scikit-learn (Vectorization)
- **AI Models:** Llama 3.3 via Dual API Cloud Nodes



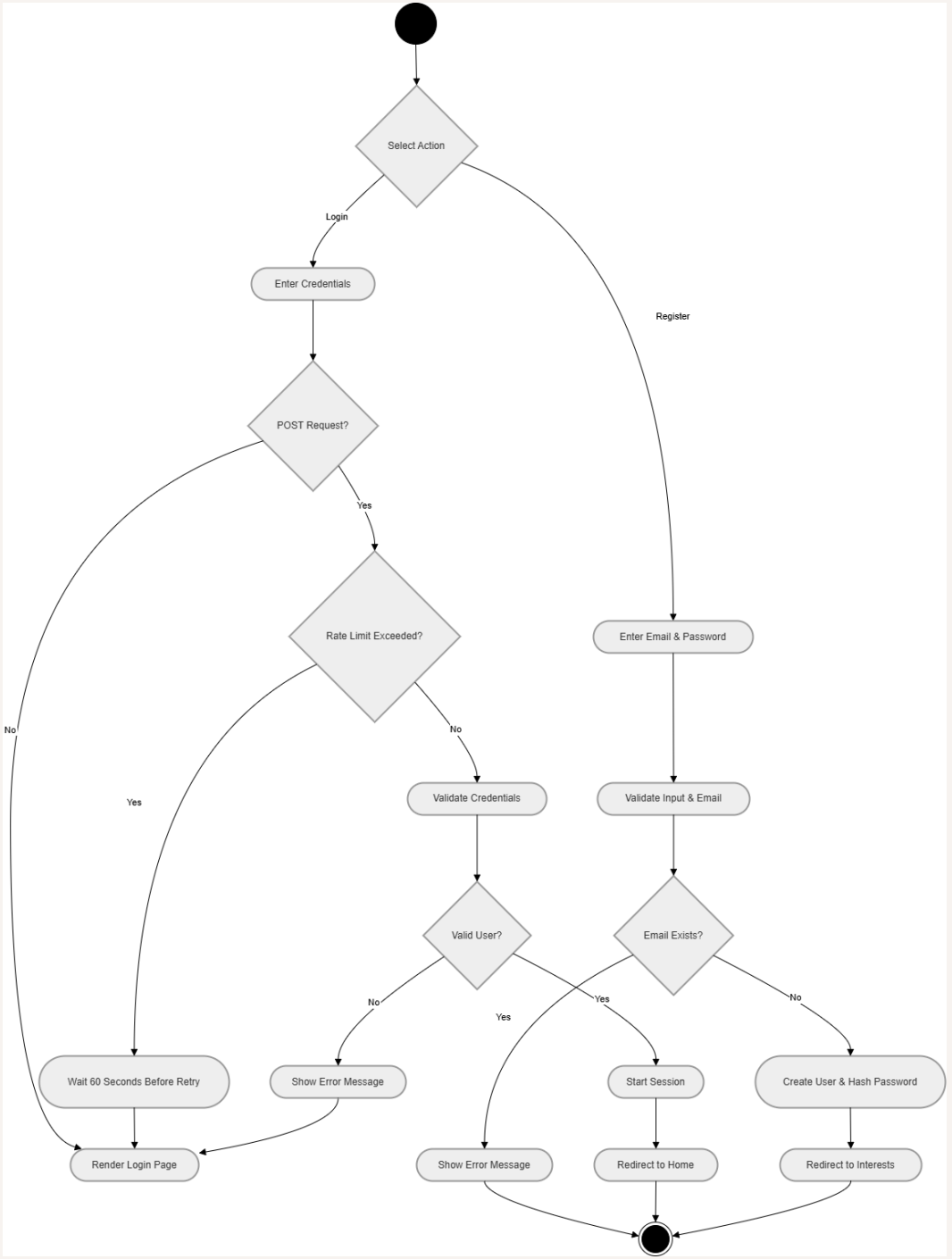
٧. Sequence Diagram



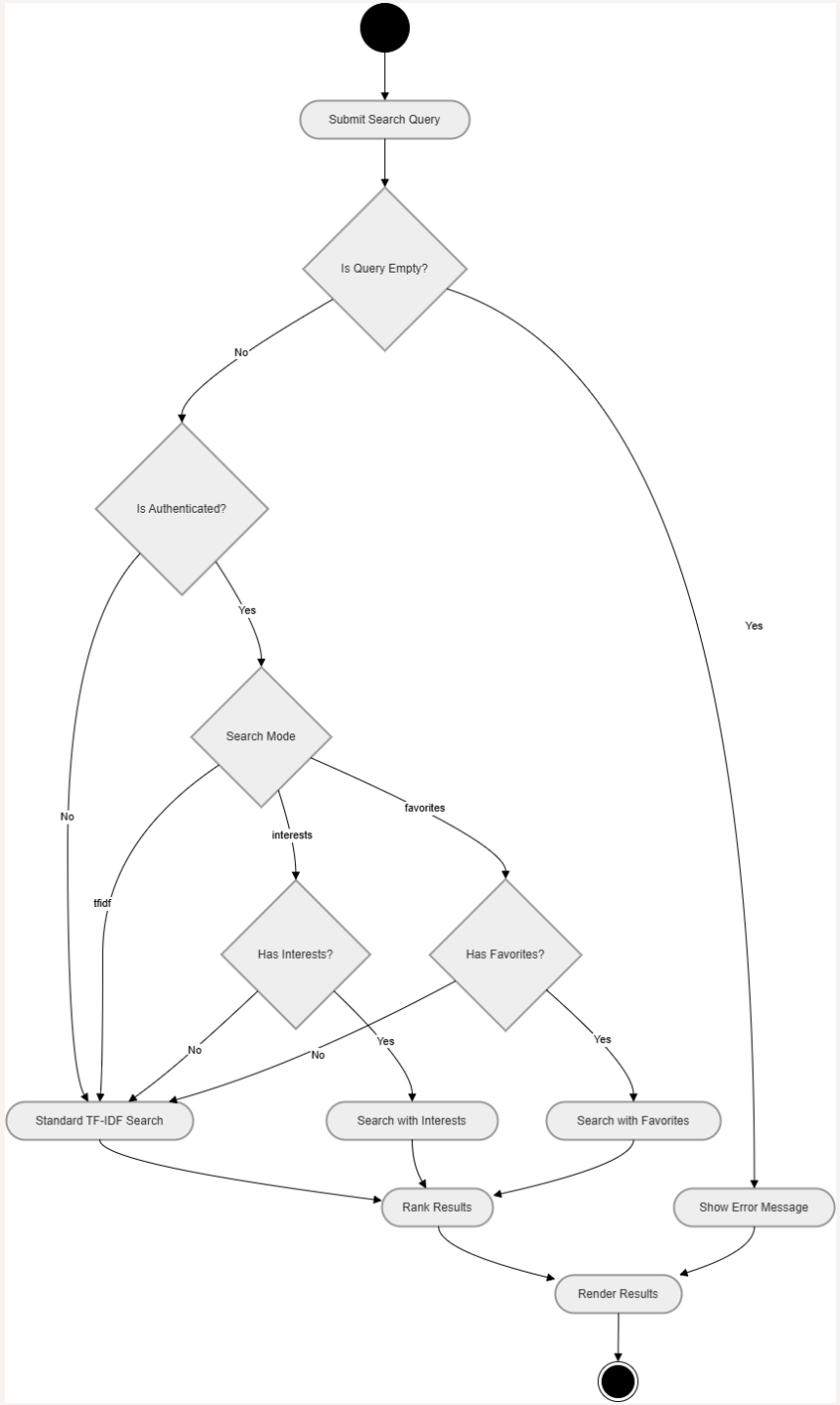
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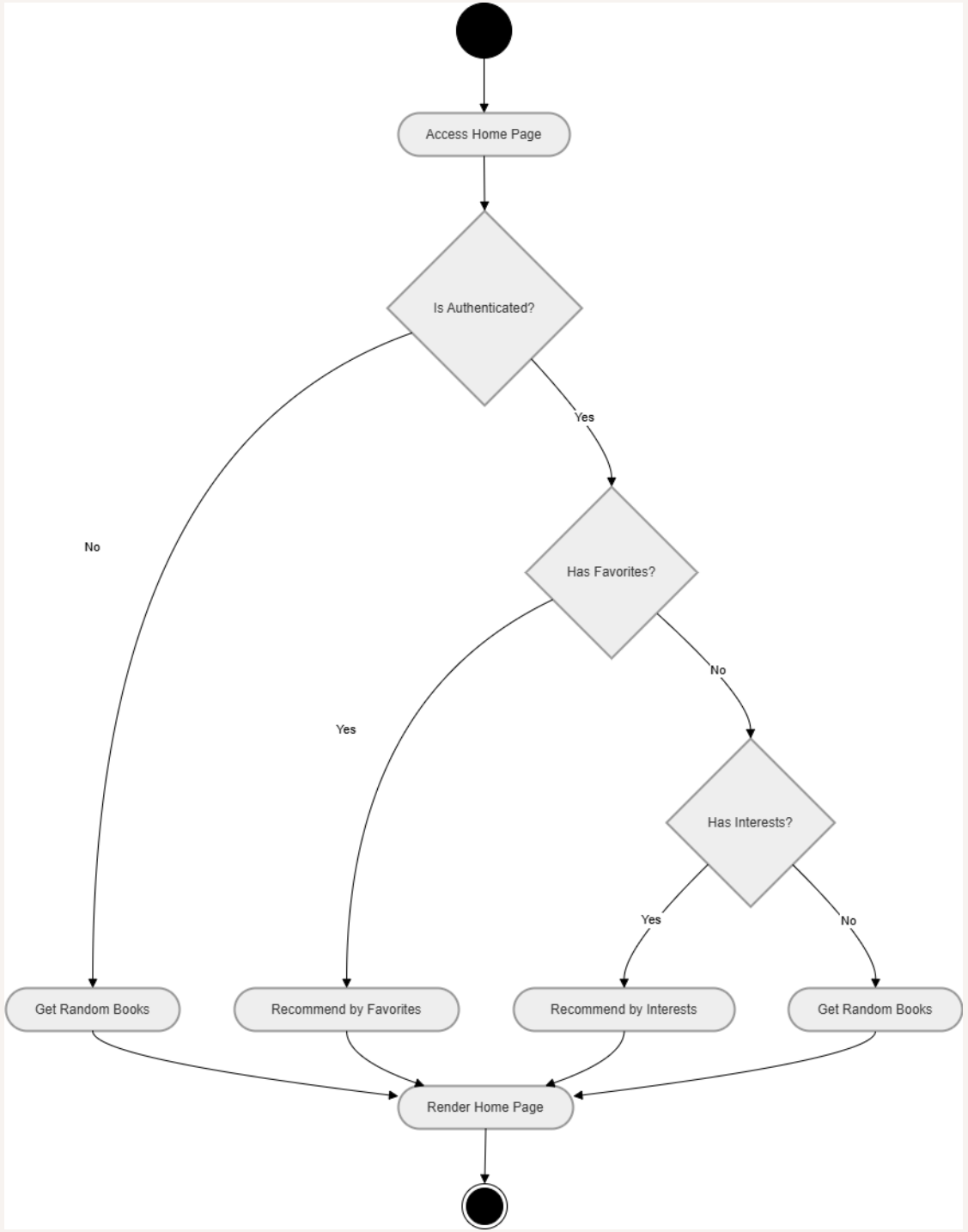
٨. Process Logic: Auth & Search Pipelines



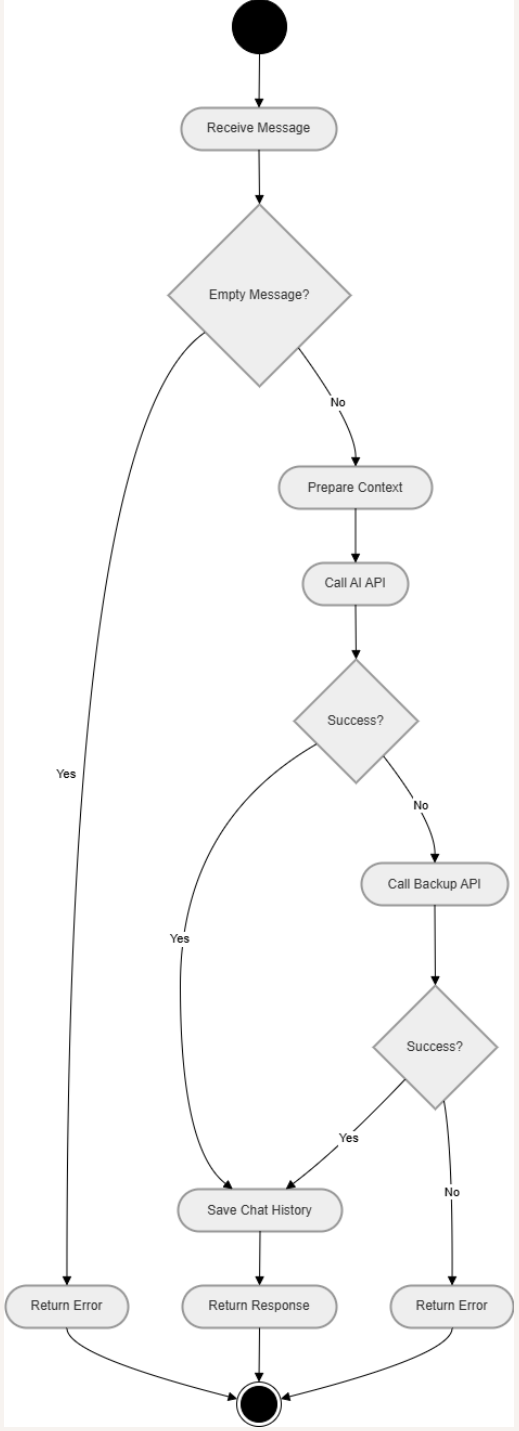
٨. Process Logic: Auth & Search Pipelines



ๆ. Process Logic: Recommendation



ๆ. Process Logic: Chatbot



١٠. Data Collection & Preprocessing

1

- **Data Collection:** The initial dataset was extracted from the Open Library API. Because of the high volume of requests across more than ٨٠ academic sub-disciplines, concurrent Python scripts were used to pull the data efficiently .

2

- **Data Cleaning:** The raw data was converted into structured tables using pandas drowyek ‹cimedaca yltcirts deniamer mrotfalp eht erusne oT . ekil sdrow gniniatnoc tnetnoc cimedaca-non evomer ot deilppa saw gniretlfi ."yretsym" ro "‹noticfi" "‹levon"

3

Data Quality Improvement (Handling Missing Data):
Combined the book's title, subjects, and tags to fill in missing descriptions.
Used Google Books API, which increased available descriptions by .٢٠%

١١. Core System Implementation: Two-Phase Architecture

- **Phase ١: Ofline training:**

Pre-computing TF-IDF matrices for١٠,٠٠٠ vocabulary tokens based on book metadata (Titles, Authors, Descriptions, Categories, Tags).

- **Phase 2: Real-Time Inference**

Executing Cosine Similarity instantly in RAM using pre-loaded weights, completely avoiding database delays.

$$w_{t,d} = TF_{t,d} \times \log \frac{N}{DF_t}$$

$$\text{Similarity}(A, B) = \frac{A \cdot B}{||A||\ ||B||} = \frac{\sum_{i=1}^{n} A_i . B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}$$

١٢. Improvement: NLP Query Resilience

SymSpell Error Correction:
Rectifies typographical errors in under 0.01ms.
Example: Maps "Artficial" ➔ "Artificial" before vector processing.

Acronym Query Expansion:
Maps technical shortcuts (e.g., 'AI', 'NLP') to full linguistic expressions. Drastically increases token match coverage inside book metadata.

١٢. Improvement: Mathematical Model - Personalized Boosting



Dynamically adjusting the base Cosine Similarity score to inject user profile context without relying on collaborative data.

$$Boost_{Category} = \beta \times 0.3 \quad (\textit{if category matches})$$

$$Boost_{Tag} = \beta \times 0.5 \quad (\textit{if tag matches via Regex})$$

$$Score_{Personalized} = Score_{Base} + Boost_{Category} + Boost_{Tag}$$

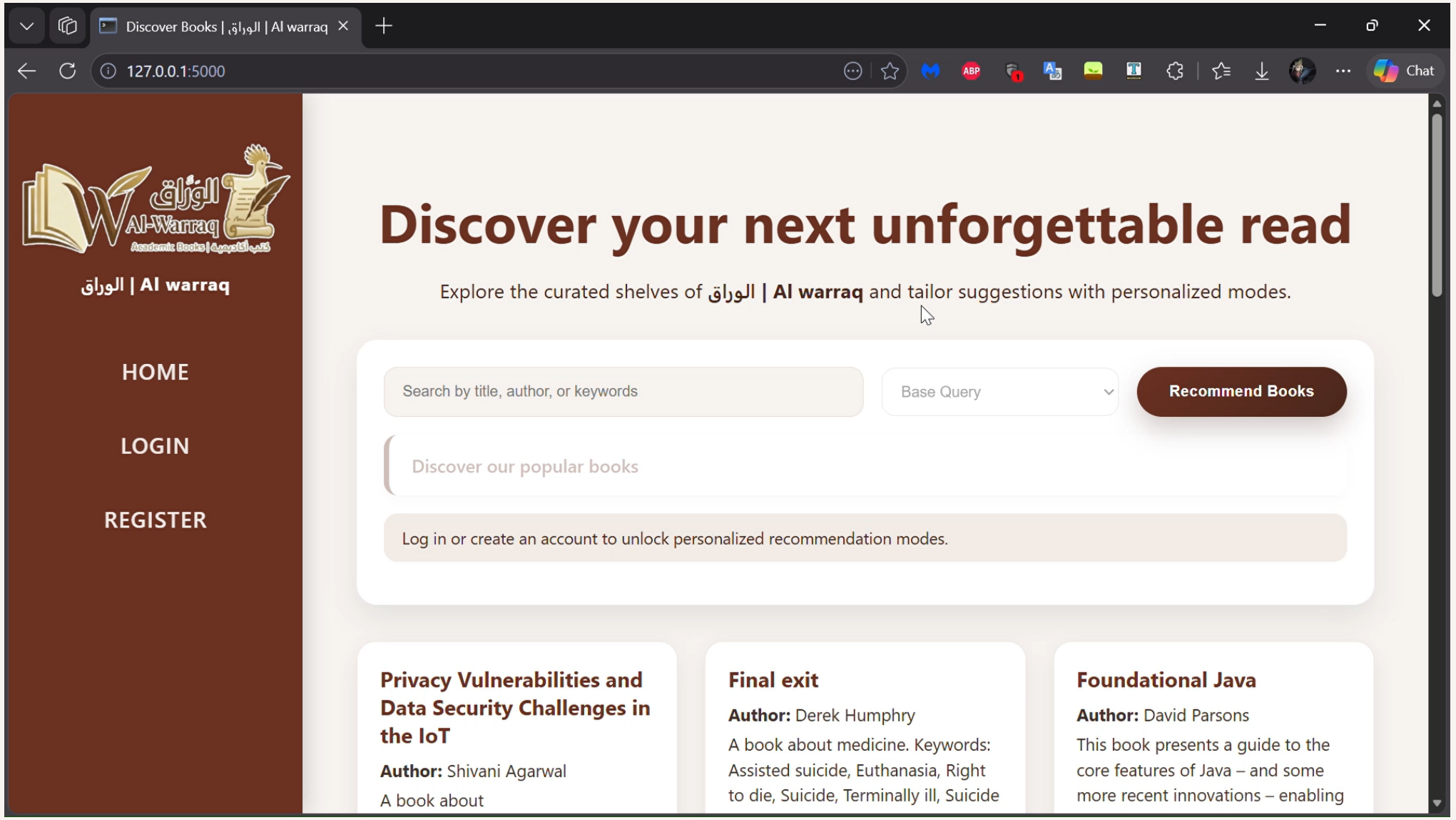
١٢. Improvement: Multi-Cloud AI Chatbot



Dual-Node Framework:
Primary: Groq Cloud API (Ultra-low latency).
Fallback: SambaNova API (Instantly takes over during network drops or quota limits).

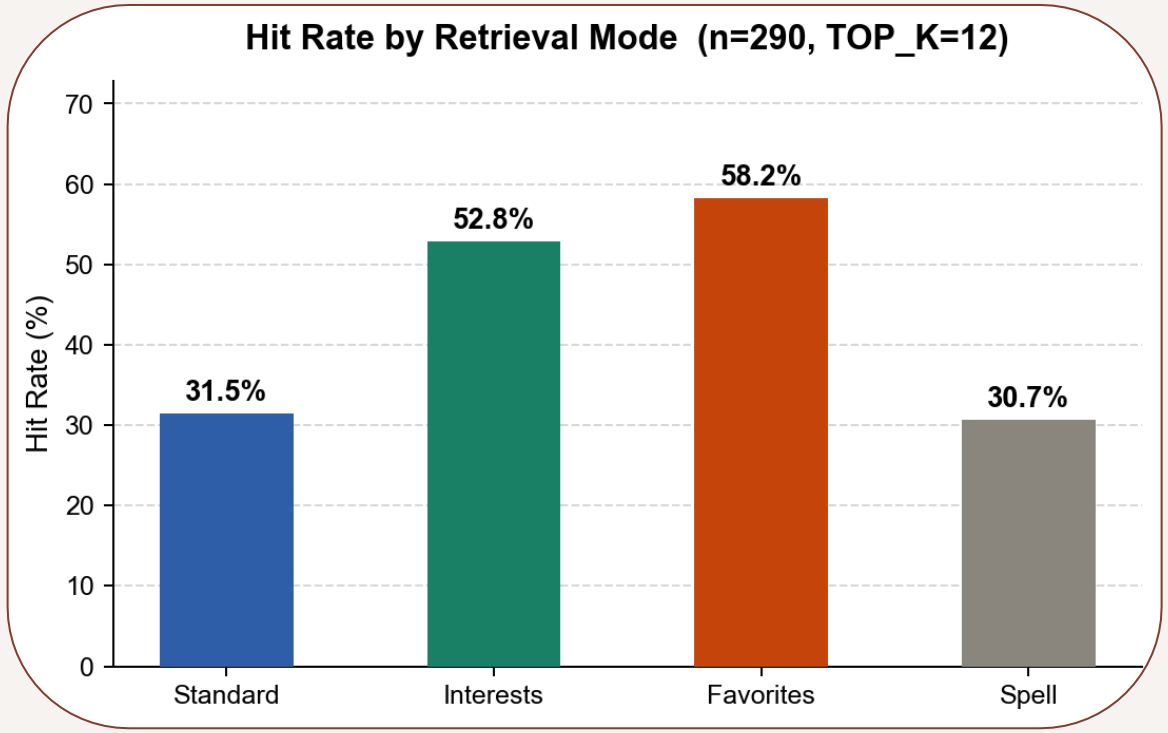
Factual Grounding:
Injects exact database book metadata into the system prompt. reduces the risk of hallucinations and adversarial prompt injections.

١٣. Demo video



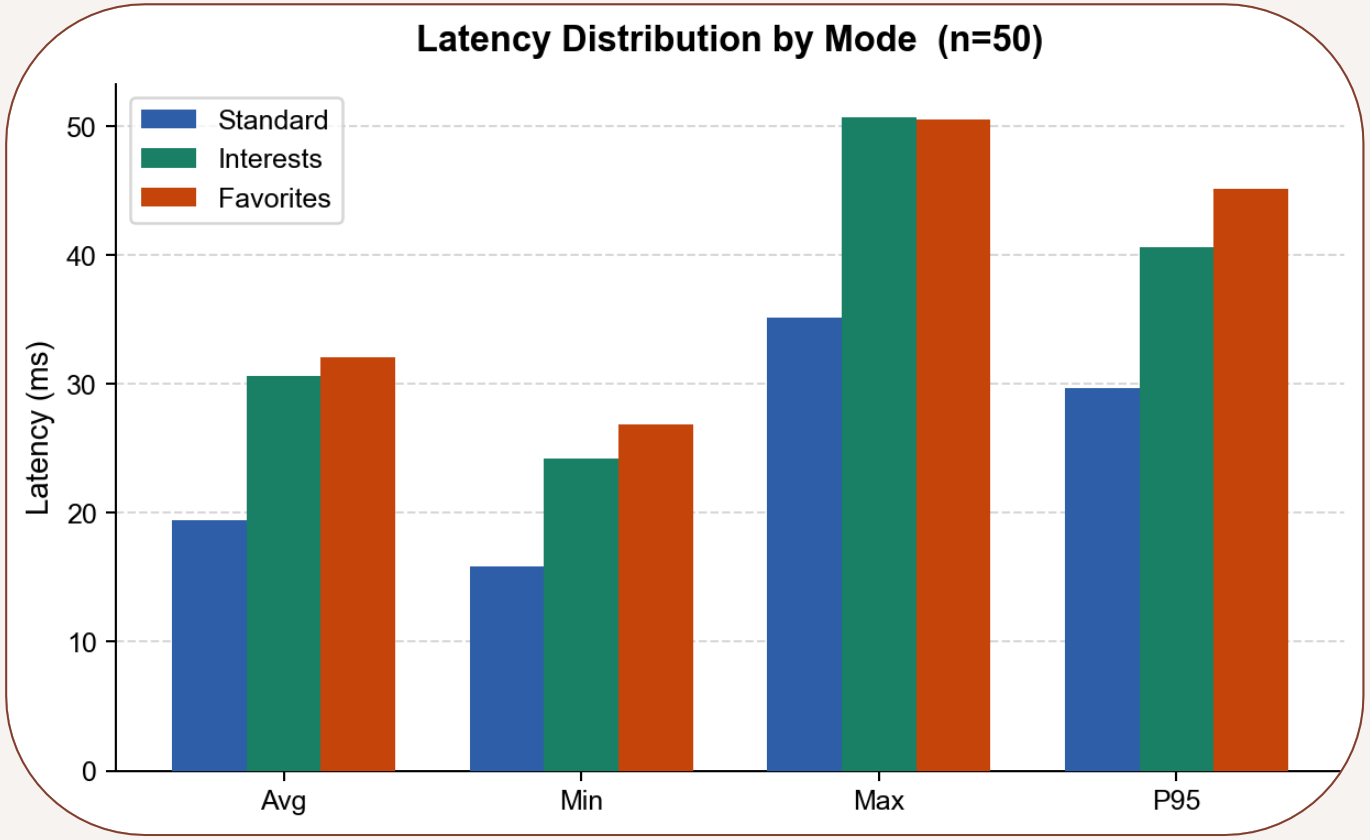
14. Evaluation: Personalization Accuracy Lift

- **Empirical Validation (290 Queries):**
- **Standard Mode:** 31.5% Hit Rate.
- **Interests-Based Boost:** 52.8% Hit Rate (+21.3% lift).
- **Favorites-Based Re-ranking:** 58.2% Hit Rate (+26.8% lift).



14. Evaluation: Personalization Accuracy Lift

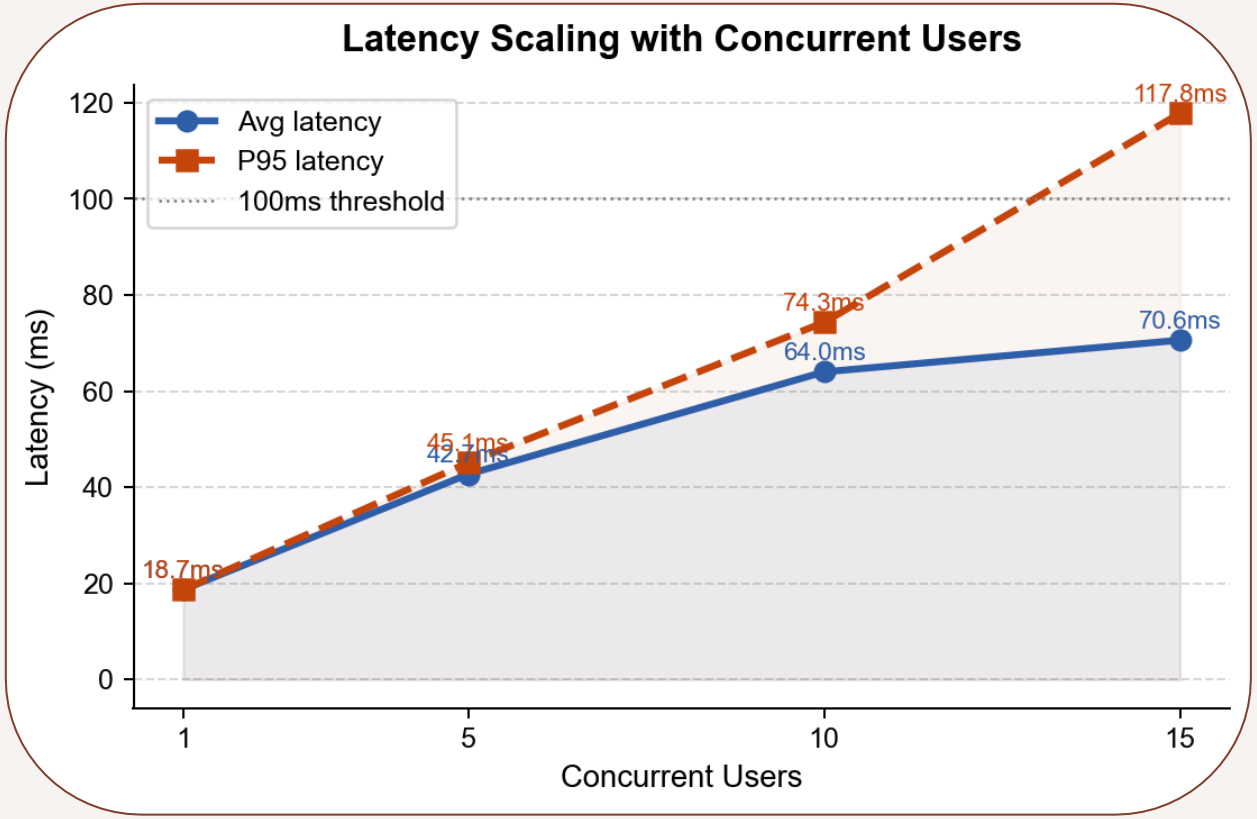
- The evaluation utilized a TOP_K parameter of 12 and measured Accuracy (Hit Rate, NDCG, Precision@1) and Speed (Average Latency).
- **Conclusion:** Persistent profile metrics dramatically increase recommendation relevance.



14. Evaluation: Concurrency & Latency

Stress Test Benchmarks:

- **1 Active User:** 18.7ms average response time.
- **15 Concurrent Users (Peak Load):** 70.6ms response time.
- **Conclusion:** Latency scales safely and efficiently well below the strict 150ms NFR boundary.



1º. Conclusion & Future Roadmap

Hybrid Models: Integrate User-to-User Collaborative Filtering layers.

Deep RAG Upgrade: Transition the Chatbot to analyze specific book chapters (Retrieval-Augmented Generation).

Campus Sync: Direct database linkage with the physical University Library inventory.

Thank you for your attention

We hope this project reflects the skills and knowledge we developed throughout our academic journey.

Any Questions?

